Automated Resume Screening and Skill Gap Analysis Using NLP

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***Abstract*—The modern recruitment landscape is characterized by an overwhelming volume of job applications, making manual resume screening both time-consuming and highly susceptible to human bias. Recruiters are often forced to make rapid judgments based on surface-level keyword matches, which leads to highly qualified candidates being overlooked while unsuitable ones are advanced. This paper proposes a comprehensive automated resume screening and skill gap analysis system based on natural language processing (NLP) techniques to address these fundamental shortcomings. Transformer-based models, specifically Sentence-BERT, is employed to extract relevant skills from unstructured text, generate dense semantic embeddings, and compute fine-grained relevance scores between candidate resumes and target job descriptions. Beyond simple matching, the system performs skill gap analysis by identifying competencies present in the job description but absent from the candidate's profile, delivering structured, personalized feedback. The system was evaluated on a curated dataset of over 13,000 resume–job description pairs labeled across the matching score. Experimental results demonstrate that the Base Model achieves a Macro F1-score of 0.4043, outperforming both fine-tuned regularized (F1 = 0.3996) and non-regularized (F1 = 0.3739) variants under single-epoch training constraints. The findings confirm measurably improved accuracy and efficiency over traditional keyword-based approaches, validating the system's viability for real-world recruitment automation. The complete source code and experimental pipeline are publicly available at https://github.com/bpragatirao/Automated-resume-screening.**

***Index Terms*—Resume screening, skill gap analysis, natural language processing, Sentence-BERT, semantic similarity, transformer models, job description matching, recruitment automation, L2 regularization.**

I Introduction

The rapid digitization of the hiring process has led to an exponential surge in the volume of job applications submitted to organizations across industries. Large enterprises routinely receive thousands of applications for a single open position, creating an enormous operational bottleneck for human resource teams. Traditional resume screening methodologies predominantly depend on keyword-based filtering, a process wherein resumes are scanned for the presence or absence of specific words or phrases nominated by a recruiter. While computationally simple, this approach is fundamentally limited: it fails to capture semantic relevance, cannot accommodate synonyms or paraphrased skill descriptions, and introduces significant human bias into the shortlisting process. A candidate who describes their expertise in "machine learning" may be incorrectly filtered out for a role that lists "deep learning" as a requirement, despite the substantial conceptual overlap between the two terms.

This work directly addresses these limitations by leveraging state-of-the-art NLP and deep learning techniques to automate resume screening and perform intelligent skill gap identification. The proposed system moves beyond lexical matching by embedding both resumes and job descriptions into a common semantic vector space, enabling nuanced, meaning-aware comparisons that are robust to vocabulary variation. This approach mirrors the cognitive process of an expert human recruiter who evaluates a candidate's potential beyond a superficial checklist of exact keywords, considering instead the broader constellation of competencies, experiences, and potential that a candidate brings.

The use of transformer-based architectures, particularly BERT [1] and its sentence-level extension Sentence-BERT [2], has marked a paradigm shift in text similarity and information retrieval. BERT's bidirectional self-attention mechanism encodes rich contextual representations of language, while Sentence-BERT builds upon this foundation to produce fixed-length sentence embeddings directly comparable using cosine distance. Applied to recruitment, these models enable a more nuanced, context-aware comparison between candidate profiles and job requirements, substantially improving both the quality and fairness of automated screening. The complete source code is publicly available at https://github.com/bpragatirao/Automated-resume-screening.

II Problem Statement

Manual resume screening at scale is fundamentally inefficient, inherently subjective, and not scalable to the demands of modern recruitment workflows. Human reviewers are constrained by the number of resumes they can meaningfully evaluate per unit of time, and fatigue-induced inconsistency further degrades the quality of assessments made late in a review session. Beyond throughput limitations, human reviewers are susceptible to a wide range of well-documented cognitive biases including affinity bias, confirmation bias, and name-based discrimination, all of which compromise the fairness of the shortlisting process and expose organizations to legal and reputational risk.

Furthermore, the absence of a standardized, quantitative matching framework means that two equally qualified candidates may receive drastically different outcomes depending solely on the reviewer assigned to their application or the time of day the review occurs. There is a clear and pressing need

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for an intelligent, automated system that can consistently and objectively match candidate resumes against job descriptions, quantify the degree of fit on a continuous scale, and identify specific skill deficits that prevent a candidate from fully meeting the requirements of a given role. Such a system would not only increase operational efficiency but also provide a fairer, more transparent, and auditable shortlisting mechanism for both recruiters and applicants.

III Objectives

The primary objectives of this work are carefully designed to address the multifaceted challenges of automated recruitment screening:

- **Automate resume screening:** Develop an end-to-end NLP pipeline capable of processing raw resume and job description text to produce ranked candidate suitability scores without manual intervention.
- **Extract and normalize skills:** Identify and standardize skill mentions from both resumes and job descriptions, handling variations in phrasing, abbreviations, and domain-specific terminology to ensure consistent representation across documents.
- **Compute semantic similarity:** Employ transformer-based sentence embeddings to calculate a meaningful, context-aware similarity score between each resume–job description pair, going substantially beyond simple token overlap.
- **Perform skill gap analysis:** Systematically identify competencies listed as requirements in a job description that are absent or insufficiently evidenced in a candidate's resume, and present these gaps as structured, actionable feedback that can guide upskilling decisions.
- **Evaluate and benchmark:** Rigorously assess the performance of multiple model variants using standard classification metrics on a labeled held-out dataset, establishing a quantitative baseline for future improvements.

IV Scope

The system in its current form is designed primarily for IT and software engineering job roles, a domain characterized by a relatively well-defined and enumerable skill vocabulary. This domain choice simplifies the skill normalization challenge and allows the model to leverage the rich pre-training of JobBERT-v2 [3], which was trained extensively on technical job postings spanning a wide range of software engineering specializations. The pipeline accepts free-form text input, meaning it does not require resumes to follow any particular structural template, increasing its practical applicability to real-world scenarios where candidates submit documents in widely varying layouts and formats.

While the initial implementation is domain-constrained, the underlying methodology and model architecture are fundamentally domain-agnostic. With collection of an appropriate labeled dataset and targeted fine-tuning, the pipeline can be extended to verticals such as healthcare, finance, legal services, or creative industries. The modular design of the system with distinct, independently testable components for skill extraction, embedding generation, similarity computation, and gap analysis, facilitates this extensibility and allows each module to be independently updated or replaced as superior models become available, ensuring the system remains relevant as the NLP landscape continues to evolve.

V State-of-the-Art Models

Our model selection was guided by a progressive review of related work, where each reference contributed a useful idea but also revealed a specific gap that motivated the next step. Table I summarises the five key works examined, the contribution each offered, and the limitation that prevented us from adopting it as our final solution.

TABLE I
SUMMARY OF REFERENCED PRIOR WORK AND DESIGN DECISIONS

Reference	Contribution	Limitation / Our Decision
Rule-based scraper [10]	Text extraction pipeline; preprocessing heuristics	No semantics; exact-string skill matching only. *Adopted preprocessing, rejected matching logic.*
Automated essay scoring [11]	Sentence-level BERT pooling for document assessment	Holistic scoring collapses skill granularity. *Adopted pooling idea; rejected single-score framing.*
NLP in compliance [12]	Domain fine-tuning outperforms frozen general BERT	Closed-label taxonomy; no open-ended skill space. *Adopted domain-adaptation insight; sought recruitment-specific model.*
Resume quality assessment [13]	BERT embeddings discriminate resume quality; multi-section encoding	Scores resumes in isolation — no JD comparison. *Confirmed BERT viability; shifted to relational matching.*
Resume ranking with cosine similarity [14]	Pairwise JD–resume ranking via cosine distance	TF-IDF vectors are lexical, not semantic; synonym-blind. *Adopted cosine ranking framework; replaced TF-IDF with semantic embeddings.*

The rule-based scraper [10] gave us a clean text-extraction pipeline but matched skills only by exact string lookup, making it blind to synonyms. The essay scoring work [11] showed that BERT sentence grouping produces useful document-level representations for assessment tasks, although its holistic scoring approach discards the skill-level granularity our gap analysis requires. The compliance study [12] confirmed that domain-adaptive fine-tuning consistently outperforms frozen general-purpose checkpoints, a key motivation for seeking a recruitment-specific base model. The IEEE resume quality paper [13] validated BERT embeddings for candidate profiling but evaluated resumes in isolation, without any reference to a target job description. Finally, the cosine-similarity ranking study [14] provided the precise relational framework we needed to classify candidates by the cosine distance between their resume vector and the job description vector, yet its TF-IDF representations were lexical and synonym-blind, confirming that the representation, not the metric, was the bottleneck.

V-A Chosen Architecture: Sentence-BERT with JobBERT-v2

These findings converged on a two-part design decision. First, we adopt **Sentence-BERT** [2] as the encoding strategy. Unlike raw BERT, which requires an $\mathcal{O}(n^2)$ cross-encoder pass for every resume–JD pair, Sentence-BERT uses a Siamese network trained with a contrastive objective to produce fixed-length sentence embeddings comparable in $\mathcal{O}(1)$ time via cosine distance. Second, rather than a general-purpose Sentence-BERT checkpoint, we use **JobBERT-v2** [5] as the backbone, a transformer encoder pre-trained on over 5.5 million job title–skill pairs from real-world job advertisements, publicly available at https://huggingface.co/TechWolf/JobBERT-v2. Its embeddings natively encode the vocabulary, co-occurrence patterns, and semantic relationships specific to the labor market domain, providing precisely the domain-specific prior that all preceding approaches lacked. Fine-tuning this model on our labeled resume–JD pairs combines domain pre-training with task-specific optimization, yielding the architecture at the core of this project.

VI Research Gaps and Motivation

Despite the impressive capabilities of BERT and Sentence-BERT in general NLP benchmarks, their direct application to recruitment screening reveals a number of significant limitations that constitute active and important areas of ongoing research.

VI-A Contextual Understanding

Although BERT is effective at capturing sentence-level semantic context through its self-attention mechanism, it may fail to correctly resolve broader document-level contextual meanings that are essential in recruitment scenarios. For example, the term “Python” may refer to the programming language in a software engineering context, or to the biological species in a zoology research posting. Without strong domain grounding, BERT-based models may conflate these distinct meanings, leading to erroneous skill extractions and incorrect match scores. Furthermore, implicit skills those not explicitly stated in a resume but strongly implied by a job title or a set of described responsibilities are frequently missed by models that rely on direct semantic overlap between the resume and job description texts.

VI-B Domain-Specific Knowledge

General-purpose pre-trained language models are trained on broad web corpora such as Wikipedia and BookCorpus, which do not adequately represent the specialized vocabulary, acronyms, and terminological conventions prevalent in technical recruitment. A phrase such as “CI/CD pipeline orchestration with Jenkins” or “microservices architecture leveraging Kubernetes and Docker” may be poorly understood by a model lacking sufficient exposure to software engineering discourse. Domain-adaptive pre-training or targeted fine-tuning on large collections of job postings and technical resumes is therefore essential for achieving reliable performance in this specific application domain a challenge that requires both significant labeled data and substantial computational investment not always available in academic research settings.

VI-C Scalability and Efficiency

Standard BERT-based models contain 110 million or more learnable parameters and require GPU acceleration for practical inference speeds. While Sentence-BERT’s pre-computation strategy mitigates the pairwise comparison bottleneck, the initial encoding of a large corpus of resumes remains computationally demanding. For organizations processing millions of applications per year, even small per-document encoding latencies accumulate into meaningful operational costs. There is therefore a compelling need for knowledge-distilled or otherwise compressed model variants that preserve the semantic quality of full-scale transformers while dramatically reducing inference time and hardware requirements, enabling cost-effective deployment on commodity infrastructure.

VI-D Bias and Fairness

Machine learning models trained on historical recruitment data inevitably encode and amplify the biases present in that data [4]. If past hiring decisions reflected systematic preferences for candidates from particular universities, demographic backgrounds, or with certain name patterns, a model trained to predict those decisions will perpetuate and scale those biases at a speed and volume that human review alone could never achieve. This is not merely a technical shortcoming but an ethical and legal concern, particularly in jurisdictions with robust anti-discrimination employment law. Addressing bias requires both careful dataset curation and the development of fairness-aware training objectives that explicitly penalize discriminatory decision boundaries.

VI-E Explainability

The black-box nature of deep neural networks is a well-recognized barrier to their adoption in high-stakes decision-making contexts. In recruitment, a candidate who has been rejected by an automated system has a legitimate interest in understanding the basis of that decision. Similarly, a recruiter who relies on an AI-generated shortlist must be able to audit the model’s reasoning for any individual recommendation. Current BERT-based models provide little to no native interpretability; while attention weights are sometimes used as proxy explanations, research has consistently shown that they do not reliably correspond to true feature importance. Post-hoc interpretation techniques such as SHAP [8] and LIME [9] are promising directions but add computational overhead to the inference pipeline and require careful validation.

VII Motivation for Automated Semantic-Based Screening

The confluence of limitations described above provides strong motivation for the development of an automated, semantics-driven screening system that transcends the capabilities of current keyword-matching tools. A semantic-based system is fundamentally better aligned with the nature of the resume screening task because it seeks to understand the *intent and content* of both the job description and the candidate's profile, rather than merely tallying lexical co-occurrences. By encoding both documents in a shared semantic space, the system can identify conceptual matches even when the precise vocabulary differs a critical capability in a domain where the same skill may be articulated in dozens of different ways across different candidates, industries, and cultural backgrounds.

Moreover, the integration of a structured skill gap analysis component transforms the system from a binary classifier into a comprehensive diagnostic tool. Rather than simply labeling a candidate as fit or non-fit, the system enumerates which specific competencies are missing from the candidate's profile relative to the job requirements, providing a foundation for targeted upskilling recommendations, personalized candidate coaching, and more transparent communication between recruiters and applicants. This shift from opaque, decision-only filtering to transparent, evidence-based evaluation is a critical step toward building recruitment AI that is not only accurate and efficient but also trustworthy and accountable.

In summary, while transformer-based models like BERT and Sentence-BERT provide a powerful and well-validated foundation for semantic text matching, critical open problems remain in domain adaptation, computational efficiency, algorithmic fairness, and model interpretability. Systematically addressing these research gaps will drive the development of the next generation of automated resume screening systems, ones that are accurate enough to be relied upon, efficient enough to be deployed at scale, fair enough to be trusted, and transparent enough to be audited by all stakeholders.

VIII Dataset Analysis

VIII-A Dataset Overview

Two datasets are used: (i) a continuous similarity dataset and (ii) a categorical similarity dataset.

a) Continuous Similarity Dataset

TABLE II
CONTINUOUS DATASET SUMMARY

Component	Value
Total pairs	9,500
Train / Validation	80 / 20
Evaluation subset	2,000 samples
Label type	Continuous score

1

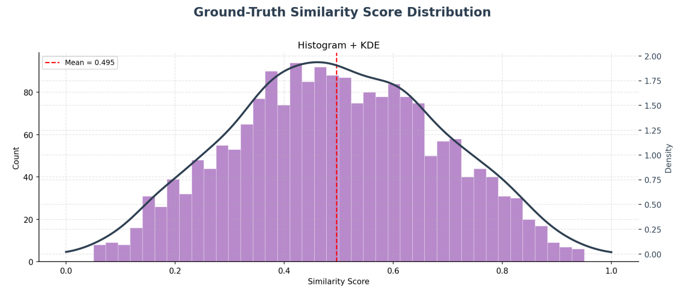

Fig. 1. **Continuous similarity score distribution**

b) Categorical Dataset.

TABLE III
LABEL DEFINITIONS AND SCORE RANGES

Label	Meaning	Score Range
No Fit	Incompatible	0.0 – 0.40
Potential Fit	Partial alignment	0.40 – 0.70
Good Fit	Strong alignment	0.70 – 1.00

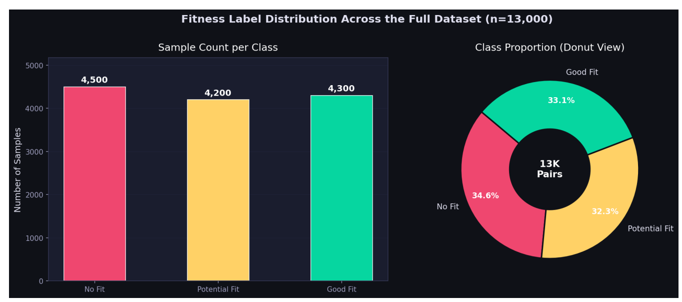

Fig. 2. **Categorical dataset distribution (13,000+ pairs).**

IX Methodology, Approach and Architecture

IX-A System Architecture Overview

The overall pipeline follows a four-stage flow: (1) raw text ingestion and preprocessing, (2) semantic encoding via the fine-tuned JobBERT-v2 backbone, (3) cosine similarity scoring, and (4) fitness classification and skill gap reporting. Figure 3 illustrates this end-to-end architecture.

IX-B Model Training Architecture

Two fine-tuned model variants were trained on the labeled dataset: a regularized model (L2 weight decay) and a non-regularized baseline. Both variants share the Siamese encoder structure illustrated in Figure 4, where identical JobBERT-v2 weights encode the resume and job description in parallel before the embeddings are compared via cosine similarity loss.

The base model used is JobBERT-v2 [5], pre-trained on over 5 million job title–skill pairs. For fine-tuning, a curated dataset of over 13,000 labeled entries was split 80/10/10 (train/validation/test) using stratified sampling. Both variants were trained for one epoch with batch size 16 and a cosine similarity objective; the regularized variant additionally applies L2 weight decay to the encoder parameters.

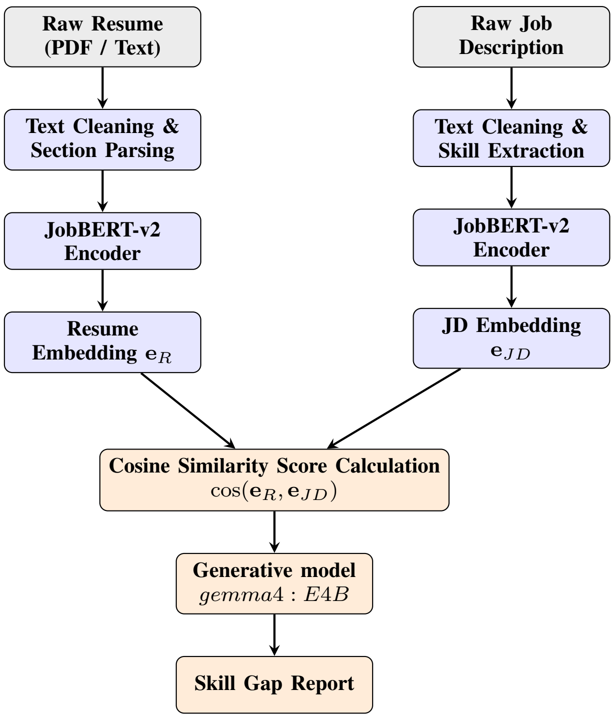

Fig. 3. End-to-end system pipeline. Both the resume and the job description are independently encoded by the fine-tuned JobBERT-v2 encoder. Cosine similarity between the two embeddings yields a fitness score. Skill gap analysis is performed by comparing extracted skill sets using generative model "gemma:7b".

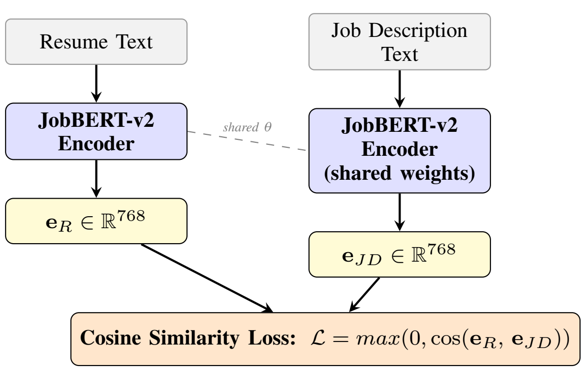

Fig. 4. Siamese encoder training architecture. Both input documents are passed through identical (weight-shared) JobBERT-v2 encoders. The cosine similarity between the resulting 768-dimensional embeddings is maximised for matched pairs and minimised for non-matched pairs during fine-tuning.

IX-C Loss with L2 Regularization

Regularization is a foundational technique in supervised machine learning that constrains the complexity of a learned model to prevent overfitting to training data and improve generalization to previously unseen examples. L2 regularization, also referred to as Ridge regularization or weight decay, achieves this by augmenting the primary training loss with a penalty term proportional to the squared Euclidean norm of the model's weight vector. The complete regularized objective is:

$$L(\mathbf{w}) = \frac{1}{n}\sum_{i=1}^{n}\ell(y_i, \hat{y}_i) + \lambda\|\mathbf{w}\|_2^2 \tag{1}$$

where $\ell(y_i, \hat{y}_i)$ is the per-sample loss between expected score label y_i and model prediction $\hat{y}_i$, $\lambda > 0$ is the regularization strength hyperparameter controlling the trade-off between data fit and weight magnitude penalty, and $\|\mathbf{w}\|_2^2$ is the squared L2 norm of the weight vector. In the least-squares regression setting this expands to:

$$L(\mathbf{w}) = \frac{1}{2n}\|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \frac{\lambda}{2}\|\mathbf{w}\|_2^2 \tag{2}$$

where $\|\mathbf{w}\|_2^2 = \sum_{j=1}^{d} w_j^2$, with d denoting the dimensionality of the parameter vector.

IX-D Gradient of the L2-Regularized Loss

To derive the weight update rule for gradient-based optimization, the gradient of the regularized loss is computed with respect to $\mathbf{w}$. Applying the chain rule to the expanded loss and combining the gradients of the data-fit term and the regularization penalty yields:

$$\nabla_{\mathbf{w}} L = -\frac{1}{n}\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\mathbf{w}) + \lambda\mathbf{w} \tag{3}$$

The first term $-\frac{1}{n}\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\mathbf{w})$ is the standard least-squares gradient that drives weights toward a better fit of the training data. The second term $\lambda\mathbf{w}$ is the regularization gradient that continuously penalizes large weight magnitudes, encouraging the optimizer to favor simpler, more parsimonious solutions over those that closely interpolate the training set at the cost of generalization.

IX-E Closed-Form Solution (Ridge Regression)

A particularly elegant property of L2 regularization in the linear regression setting is that the regularized objective is strictly convex and admits a unique closed-form global minimizer. Setting the gradient to zero and solving the resulting linear system:

$$\mathbf{X}^\top\mathbf{X}\mathbf{w} + n\lambda\mathbf{w} = \mathbf{X}^\top\mathbf{y} \tag{4}$$

$$(\mathbf{X}^\top\mathbf{X} + n\lambda\mathbf{I})\mathbf{w} = \mathbf{X}^\top\mathbf{y} \tag{5}$$

$$\mathbf{w}^* = (\mathbf{X}^\top\mathbf{X} + n\lambda\mathbf{I})^{-1}\mathbf{X}^\top\mathbf{y} \tag{6}$$

The addition of the $n\lambda\mathbf{I}$ term to the Gram matrix $\mathbf{X}^\top\mathbf{X}$ guarantees that the matrix to be inverted is always positive definite and therefore numerically non-singular, even when the data matrix $\mathbf{X}$ is rank-deficient — a scenario that frequently arises in high-dimensional text feature spaces where the number of features exceeds the number of training samples.

IX-F Gradient Descent Update Rule

For large-scale neural network training where closed-form inversion of the Gram matrix is computationally intractable, the regularized objective is minimized iteratively via stochastic gradient descent. Substituting the derived gradient into the standard gradient descent update equation and rearranging:

$$\mathbf{w} := \mathbf{w} - \eta\left(-\frac{1}{n}\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\mathbf{w}) + \lambda\mathbf{w}\right) \tag{7}$$

Factoring the weight term reveals the weight decay form:

$$\mathbf{w} := (1 - \eta\lambda)\mathbf{w} + \frac{\eta}{n}\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\mathbf{w}) \tag{8}$$

The multiplicative factor $(1-\eta\lambda)$ applied to $\mathbf{w}$ at each update step is the origin of the term *weight decay*: irrespective of the gradient signal, the weight vector is shrunk toward zero by a constant fraction $\eta\lambda$ at every iteration. This continuous shrinkage prevents any individual parameter from growing excessively large, encoding an inductive bias toward low-norm solutions. In modern deep learning frameworks such as PyTorch and TensorFlow, L2 regularization is typically implemented directly as a `weight_decay` argument to the optimizer, exploiting this equivalence rather than modifying the loss function explicitly.

X Experimental Results and Analysis

X-A Evaluation Protocol

A comprehensive evaluation framework was designed to assess both categorical fit classification and continuous similarity estimation. For classification, a dedicated held-out test set was constructed to ensure that reported metrics reflect genuine out-of-sample generalization rather than in-sample memorization. This test set comprises 1,000 entries sampled uniformly at random from a 6,000-record evaluation pool, with each sample annotated using three ordinal fitness categories: *No Fit*, *Potential Fit*, and *Good Fit*. Macro-averaged F1-score was selected as the primary evaluation metric, as it assigns equal importance to all classes regardless of their distribution, providing an unbiased assessment in an imbalanced multi-class setting.

In parallel, a second dataset consisting of 9,500 resume–job description pairs annotated with continuous similarity scores was used to evaluate fine-grained semantic alignment. From this dataset, 2,000 samples were drawn uniformly at random and evaluated against their labeled similarity scores. Unlike the categorical setup, this evaluation focuses on the agreement between predicted and true similarity values, providing a complementary assessment of performance in a continuous scoring setting

X-B Results and Discussion

a) Categorical Evaluation.

The performance of all three model variants on the classification task is summarized in Table IV. The Base Model achieves the highest Macro F1-score of 0.4043, indicating the strongest overall balance across the three classes. Its confusion matrix shows relatively strong identification of the *No Fit* category, while confusion persists between *Potential Fit* and *Good Fit*, reflecting the inherent ambiguity between partially and strongly matching candidates.

The Regularized Model achieves a comparable Macro F1-score of 0.3996 but exhibits a systematic bias toward predicting *Good Fit*, as evidenced by the concentration of predictions in the corresponding column. This suggests reduced class discrimination despite similar aggregate performance. The Non-Regularized Model performs worst (Macro F1 = 0.3739),

TABLE IV
ML MODEL EVALUATION ON JD–RESUME SIMILARITY DATASET
($N = 1000$)

Model	Macro F1	Class	Confusion Row
Base Model	0.4043	No Fit	[309, 175, 22]
		Potential Fit	[85, 140, 28]
		Good Fit	[68, 144, 29]
Regularized	0.3996	No Fit	[179, 148, 156]
		Potential Fit	[21, 65, 168]
		Good Fit	[21, 72, 170]
Non-Regularized	0.3739	No Fit	[150, 156, 177]
		Potential Fit	[12, 56, 186]
		Good Fit	[18, 60, 185]

with a more pronounced collapse toward the *Good Fit* class, indicating poor generalization and limited ability to separate categories.

b) Continuous Similarity Evaluation.

Evaluation on the continuous similarity dataset indicates a clear improvement in predictive alignment for fine-tuned models compared to the base model, as summarized in Table V and illustrated in Figure 5.

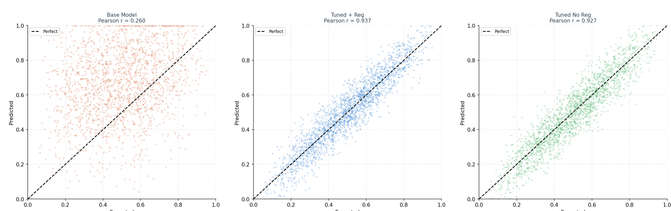

Fig. 5. **Continuous similarity evaluation on 2,000 sampled instances from the 9,500-entry dataset.** Predicted vs. Expected similarity scores across model variants.

TABLE V
CONTINUOUS EVALUATION METRICS

Metric	Base	Reg	NoReg
MAE $\downarrow$	0.271	0.076	0.078
RMSE $\downarrow$	0.326	0.098	0.099
Pearson $\uparrow$	0.298	0.807	0.806
Spearman $\uparrow$	0.281	0.788	0.784
Accuracy @ ± 0.1 $\uparrow$	0.224	0.720	0.717

XI Conclusion

This paper has presented a semantic-based resume screening system that improves upon traditional keyword-based approaches by leveraging dense vector representations to capture conceptual alignment between resumes and job descriptions. In addition to fit assessment, the system incorporates a skill gap analysis component that provides actionable insights into missing competencies, enhancing its practical utility for both recruiters and applicants.

Experimental evaluation across two complementary settings highlights distinct model behaviors. On the categorical classification task (1,000-sample test set), the Base Model achieves the highest Macro F1-score (0.4043), outperforming both the regularized (0.3996) and non-regularized (0.3739) variants, which exhibit majority-class bias and reduced class discrimination. In contrast, results on the continuous similarity dataset (2,000 samples drawn from 9,500) show that both fine-tuned models substantially outperform the Base Model, achieving significantly lower error and stronger correlation with ground-truth similarity scores. This indicates improved capability in capturing fine-grained semantic alignment despite weaker categorical separation.

These findings suggest a trade-off between discrete classification performance and continuous similarity estimation. While the Base Model retains stronger inductive biases for categorical decision boundaries under limited training, fine-tuning enables more precise semantic scoring, which is critical for ranking and matching tasks.

Several directions for future work follow from these observations. Extending the current training setup to multi-epoch fine-tuning with validation-guided early stopping may improve both classification stability and reduce bias. Evaluating on larger and more diverse datasets across domains would strengthen generalizability. Incorporating interpretability techniques can improve transparency of matching and skill gap outputs. Finally, systematic fairness evaluation and bias mitigation remain essential for responsible deployment in real-world recruitment settings. The complete implementation supporting this work is publicly available at https://github.com/bpragatirao/Automated-resume-screening.

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